# CHAPTER 2

# THEORETICAL FRAMEWORK AND REVIEW OF RELATED LITERATURE AND STUDIES

## Theoretical Framework

### Technology Acceptance Model

The Technology Acceptance Model (TAM) developed by Davis (1989) provides an appropriate theoretical foundation for the proposed **ResProg: Web-Based Research Progress Tracking and Document Management System for Senior High at Calinog National Comprehensive High School**. TAM explains how users come to accept and use an information technology based primarily on two factors: **perceived usefulness** and **perceived ease of use**. Perceived usefulness refers to the extent to which a person believes that using a particular system will improve his or her performance, while perceived ease of use refers to the extent to which a person believes that using the system will require minimal effort.

TAM is appropriate for ResProg because the proposed system is intended to support both **students and research advisers** in managing research-related activities. Students are expected to use the system to monitor their research progress, submit documents, view adviser feedback, and receive notifications regarding pending tasks and deadlines. Research advisers, on the other hand, are expected to use the system to monitor assigned research groups, access submitted documents, provide feedback, and track the completion of research requirements.

The **perceived usefulness** component of TAM is particularly relevant because ResProg is intended to reduce difficulties associated with manually monitoring research progress and handling multiple research documents. A centralized system can provide advisers with a clearer view of student progress while allowing students to access their research status and feedback without repeatedly asking advisers for routine updates.

The **perceived ease of use** component is equally important because the system will be used by senior high school students and research advisers with potentially different levels of technological familiarity. A simple interface, organized document repository, accessible progress indicators, and understandable notifications may contribute to easier system adoption.

The applicability of TAM to electronic document management in Philippine educational institutions is also supported by Aliazas, Dela Cruz, and Ilagan (2024), who examined an Electronic Document Management System in a Philippine higher education institution. Their study investigated the relationship between electronic document management and TAM components, including perceived usefulness, perceived ease of use, and behavioral intention to use the system.

Therefore, TAM provides an appropriate theoretical basis for evaluating whether ResProg is perceived by its intended users as **useful, accessible, and easy to use** in managing senior high school research activities.

## Conceptual Framework: IPO Model

The study adopts the **Input–Process–Output (IPO) Model** as its conceptual framework. The model illustrates how the resources and information required by the system are transformed through specific

processes into the intended system output.

### Input

The inputs of the proposed ResProg system include: student and adviser account information; research group information; research titles and project information; research milestones and required tasks; research documents and manuscript drafts; submission records; adviser feedback and evaluation; research deadlines; progress updates; and notification and reminder information.

### Process

The inputs undergo several processes, including user account and access management; registration and organization of research groups; assignment of research advisers; recording of research milestones; uploading and submission of research documents; monitoring of research progress; review of submitted documents; recording and displaying adviser feedback; updating research status; generation of notifications and reminders; tracking pending and completed research requirements; and management and storage of research documents.

### Output

The expected output is a functional **ResProg: Web-Based Research Progress Tracking and Document Management System for Senior High at Calinog National Comprehensive High School** that provides online research document submission, research progress monitoring, centralized document management, adviser feedback, research milestone tracking, deadline notifications and reminders, student access to research status and feedback, adviser access to submitted research documents, and improved organization of research monitoring activities.

The IPO framework is appropriate because the proposed system transforms research-related information and documents into organized progress information, feedback, and monitoring outputs. This is consistent with the provided module, where defined inputs undergo organized processes to produce the intended system output.

### Feedback Component

A feedback component may also be incorporated into the conceptual framework. Feedback from students and research advisers can be used to identify usability concerns, system issues, document-management problems, and possible improvements. The feedback can serve as a basis for system maintenance, refinement, and future enhancement.

The importance of continuous feedback is particularly relevant to ResProg because research supervision involves repeated cycles of **submission, checking, feedback, revision, and resubmission**. An online thesis supervision case demonstrated that maintaining a research log, sharing documents digitally, and providing prompt written feedback can support communication and time management while reducing the need to exchange multiple documents through email and other communication channels.

## Review of Related Literature

### Research in Senior High School Education

Research is an important component of Senior High School education because it develops learners' abilities to formulate problems, review existing knowledge, collect and analyze information, and communicate findings. The Department of Education's Senior High School Practical Research curriculum includes

competencies involving the research process, literature review, research design, data collection, analysis, reporting, and presentation of research findings.

The research process therefore involves multiple activities that must be completed sequentially. Students need to develop research topics, formulate research questions, prepare literature reviews, conduct data collection, analyze information, and prepare their final research output. Because these activities occur over an extended period, monitoring the progress of research groups can become an important administrative and instructional responsibility.

For senior high school research advisers, monitoring becomes more challenging when several groups are working simultaneously. Each group may have different levels of progress, document versions, deadlines, and revision requirements. A system that organizes these elements in one platform can provide a structured approach to research monitoring.

## Research Progress Monitoring

Research progress monitoring involves systematically determining whether students or research groups are completing their assigned research activities according to established milestones and schedules. Monitoring allows advisers to determine which research requirements have been completed, which remain pending, and which groups may require additional guidance.

Digital research monitoring can provide greater visibility into the current status of research projects. Instead of relying solely on verbal updates or manually checking multiple documents, advisers can view progress information through a centralized interface.

Lando and Wambua (2025) examined the development of a Thesis Management System intended to address problems involving delayed feedback, communication difficulties, and supervision follow-up. Their study reported that the system helped create a more structured supervision process and improved interactions between students and supervisors.

Similarly, an online thesis supervision process described by Aarhus University used a research log that students regularly updated to record their progress and reflections. Supervisors then used the log and shared documents to provide feedback and monitor the research process. The implementation demonstrated the usefulness of structured digital monitoring and regular feedback in maintaining research momentum.

These findings support the inclusion of a **research progress monitoring feature** in ResProg.

## Online Research Document Management

Document management refers to the organized storage, retrieval, processing, and maintenance of digital documents. In academic environments, document management can involve research proposals, manuscript drafts, revised chapters, evaluation documents, and other research-related files.

Traditional research supervision may require advisers to handle multiple physical or digital copies of student manuscripts. This may make it difficult to determine which document represents the latest version and whether previous feedback has already been incorporated.

Electronic Document Management Systems can address some of these challenges by centralizing documents and making them easier to access and organize. Aliazas et al. (2024) investigated electronic document management in a Philippine higher education institution and examined its application to administrative processes, academic management, student services, and research activities. Their findings

indicate the relevance of electronic document management to educational operations and its relationship with technology acceptance factors.

For ResProg, document management is not limited to simply storing files. The proposed system integrates document submission with **research progress tracking and adviser feedback**. This allows a submitted document to be associated with a specific research group, research milestone, submission status, and adviser response.

### Online Submission and Adviser Feedback

Feedback is an important component of research supervision because students often require guidance when revising their research manuscripts. Through feedback, advisers can identify areas that require correction, improvement, clarification, or further development.

Digital platforms can facilitate feedback by allowing advisers to provide comments alongside submitted documents without requiring students and advisers to be physically present at the same location.

Recent literature on virtual thesis supervision emphasizes the role of supervisor feedback in students' research development and highlights the importance of meaningful academic interaction during online supervision.

Research on online thesis supervision has also identified timely feedback as an important concern. Previous supervision research reported problems involving delayed feedback, limited communication, and difficulties in monitoring the thesis-supervision process.

For ResProg, the document submission and feedback feature is therefore designed to create a more organized cycle: **Submit** → **Review** → **Provide Feedback** → **Revise** → **Resubmit**. This cycle corresponds closely with the actual nature of research manuscript development.

### Notifications and Reminders

Notifications and reminders are important features in systems that involve deadlines, tasks, and sequential activities. Research projects frequently involve scheduled requirements such as proposal preparation, manuscript submission, revisions, presentations, and final submission.

Without an organized reminder mechanism, students may overlook deadlines or fail to recognize pending research requirements. Advisers may also need to repeatedly remind individual research groups about incomplete tasks.

The integration of notifications into ResProg is intended to reduce unnecessary manual follow-ups by providing students with information about **upcoming deadlines, pending tasks, and required submissions**.

This feature complements the progress tracking function because monitoring alone informs users about the current state of a research project, while notifications encourage users to take action when a task or deadline requires attention.

## Related Studies

### Digital Solutions for Thesis Supervision

Lando and Wambua (2025) developed and examined a Thesis Management System in response to challenges encountered in thesis supervision. The study identified delayed feedback and difficulties in following up students' thesis progress as significant concerns. The proposed system was intended to improve supervision efficiency, reduce delays, and create a more structured interaction between students and supervisors.

The study is relevant to ResProg because both systems seek to improve research supervision through digital technology. However, the proposed ResProg system differs in its target environment. While the previous study focuses on graduate-level thesis supervision, ResProg is specifically designed for **Senior High School research at Calinog National Comprehensive High School**.

### Online Thesis Management System

A 2026 study on the design and implementation of an online thesis management system examined how digital technology could address limitations of manual thesis supervision, including communication delays, lack of transparency, and difficulties in monitoring student progress. The proposed system incorporated structured system development activities and included functions for facilitating interaction between students and thesis advisers.

This study is relevant to ResProg because it demonstrates that research supervision can be supported through an integrated digital platform. Both systems emphasize progress monitoring and interaction between students and their advisers.

However, ResProg is intended for the Senior High School context and focuses on a more localized research workflow involving research groups, milestones, document submission, adviser feedback, and deadline reminders.

### Thesis Supervision Mobile System

A study on a Thesis Supervision Mobile System proposed an online tool to improve student–advisor communication during the thesis process. The system included functions related to file uploading, student and adviser accounts, and communication. Usability testing indicated positive perceptions among participating students and advisers, particularly regarding communication, document viewing, and feedback.

The study demonstrates the potential of digital systems to support communication between students and research advisers. It is particularly relevant to ResProg's document submission and feedback features.

However, ResProg expands beyond communication and file uploading by integrating **progress monitoring, milestone tracking, notifications, reminders, and document management** into one web-based platform.

### Electronic Document Management in Philippine Higher Education

Aliazas, Dela Cruz, and Ilagan (2024) studied Electronic Document Management Systems in a Philippine higher education institution. The study examined how electronic document management could support administrative processes, academic management, student services, and research activities. The researchers also examined the system through the Technology Acceptance Model.

The study is particularly relevant because it provides a Philippine educational context for the use of electronic document management. It also supports the use of TAM in evaluating technology adoption.

ResProg builds upon this concept by focusing specifically on **research documents and research-progress monitoring in Senior High School**, rather than general institutional document management.

## Related Systems

### Electronic Document Management Systems

Electronic Document Management Systems provide digital mechanisms for storing, organizing, accessing, and managing documents. In educational institutions, such systems can support academic and administrative processes by reducing dependence on physical documents and fragmented file storage.

The study by Aliazas et al. (2024) demonstrates the applicability of electronic document management within a Philippine educational institution and identifies its relevance to research-related activities.

ResProg adopts the document-management concept but integrates it with research monitoring. Therefore, a research document is not treated merely as a stored file; it becomes part of a specific research group's progress and submission history.

### Online Thesis Supervision Systems

Online thesis supervision systems provide digital environments where students and advisers can interact, exchange documents, monitor research activities, and provide feedback.

One example is an online thesis supervision platform that allows students to access thesis-related activities through an online environment and supports a structured progression from proposal development to later research stages. Such systems can provide mechanisms for students to upload revised documents and access supervisor feedback.

This type of system is highly relevant to ResProg because of its integration of **research stages, document submission, and feedback**. However, ResProg is designed for Senior High School research and is intended to address the specific research-monitoring needs of Calinog National Comprehensive High School.

## Review of Related Literature and Studies

The reviewed literature and studies consistently indicate that digital systems can support research supervision by organizing documents, improving communication, monitoring progress, and facilitating feedback. Traditional research supervision may require students and advisers to depend on physical documents, email, messaging applications, or separate file repositories. These approaches may make it difficult to maintain a clear view of research progress when several research groups are being supervised simultaneously.

The literature also demonstrates the importance of **structured progress monitoring**. Digital research logs, thesis management systems, and online supervision platforms provide mechanisms through which students can report progress while advisers can monitor research activities. The use of organized progress information can reduce the need for repeated manual inquiries and allow advisers to focus more directly on academic guidance.

Another common theme is the importance of **timely feedback**. Research supervision involves repeated cycles of submission, evaluation, revision, and resubmission. Digital platforms can support this cycle by allowing students to submit documents and advisers to provide feedback through a centralized environment.

The literature further indicates that **document management** is an important component of educational information systems. Electronic document management can improve the organization and accessibility of academic and research-related documents. Philippine research on EDMS demonstrates its applicability in higher education and its connection with technology acceptance.

The reviewed studies, however, vary in their target users and institutional settings. Many existing systems focus on university-level thesis or graduate research supervision. ResProg differs by focusing on **Senior High School research**, where students work within the requirements of the K to 12 curriculum.

## Research Gap

Although existing studies have demonstrated the usefulness of digital systems for thesis supervision, electronic document management, and online academic feedback, a gap remains in the development of a **localized web-based research progress tracking and document management system specifically designed for Senior High School research**.

Existing thesis management systems primarily target higher education and graduate-level research. Their workflows may therefore not directly correspond to the research requirements, group structure, and supervision context of Senior High School students.

Furthermore, existing electronic document management systems primarily emphasize document organization and institutional operations. They do not necessarily integrate research milestones, progress indicators, adviser feedback, submission monitoring, and deadline reminders into one research-specific platform.

ResProg addresses this gap by combining these functions into a centralized web-based system intended specifically for **Senior High School research at Calinog National Comprehensive High School**. The proposed system integrates research progress monitoring, research milestone tracking, online document submission, adviser document checking, adviser feedback, student access to research status, deadline notifications, pending-task reminders, and centralized research document management.

Thus, the proposed study does not merely digitize document storage. It seeks to create an integrated research-monitoring environment that connects **students, advisers, documents, milestones, feedback, and deadlines** within one platform.

## Synthesis

The reviewed literature and studies provide support for the development of ResProg. First, the Department of Education's Senior High School research curriculum demonstrates that research is a structured process involving several interconnected activities. Second, research supervision literature indicates that monitoring, communication, and timely feedback are important concerns in research management. Third, studies on electronic document management demonstrate the usefulness of digital systems in organizing academic and research-related documents.

The studies also demonstrate that online supervision systems can provide centralized environments for document submission, feedback, and research progress monitoring. However, most identified systems are designed for higher education or broader institutional applications.

Therefore, ResProg builds upon the established concepts of **digital document management, online research supervision, progress monitoring, feedback management, and technology acceptance**, while

contextualizing them for the Senior High School environment of Calinog National Comprehensive High School.

The use of **TAM** provides a theoretical basis for evaluating users' acceptance of the proposed system, particularly its usefulness and ease of use. The **IPO model** provides the conceptual structure for organizing the system's inputs, processes, outputs, and feedback mechanism. Together, these frameworks support the development of a system intended to make research monitoring more organized, accessible, and efficient.

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